

Use Case Diagrams | Unified Modeling Language (UML)

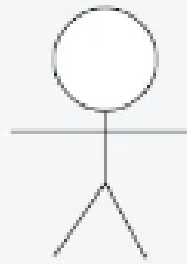
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*Lecture (Senior Scale)
SWE, DIU*

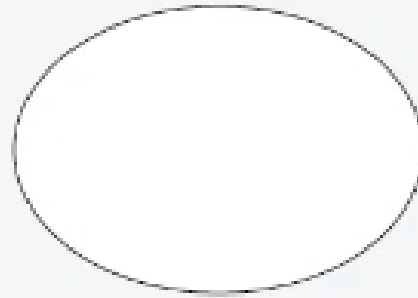
Use Case Diagram

- A Use Case Diagram is a vital tool in system design, it provides a visual representation of how users interact with a system.
- It serves as a blueprint for understanding the functional requirements of a system from a user's perspective, aiding in the communication between stakeholders and guiding the development process.

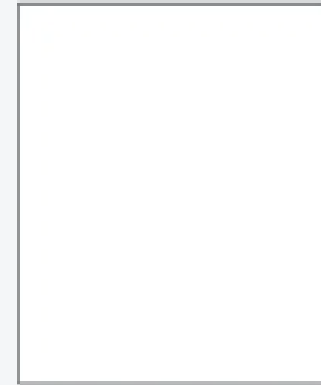
Use Case Diagram Notations



Actor



UseCase



System Boundary

Use Case Diagram Relationships

- i. Association Relationship
- ii. Include Relationship
- iii. Extend Relationship
- iv. Generalization Relationship

Association Relationship

- The Association Relationship represents a communication or interaction between an actor and a use case.
- It is depicted by a line connecting the actor to the use case.
- This relationship signifies that the actor is involved in the functionality described by the use case

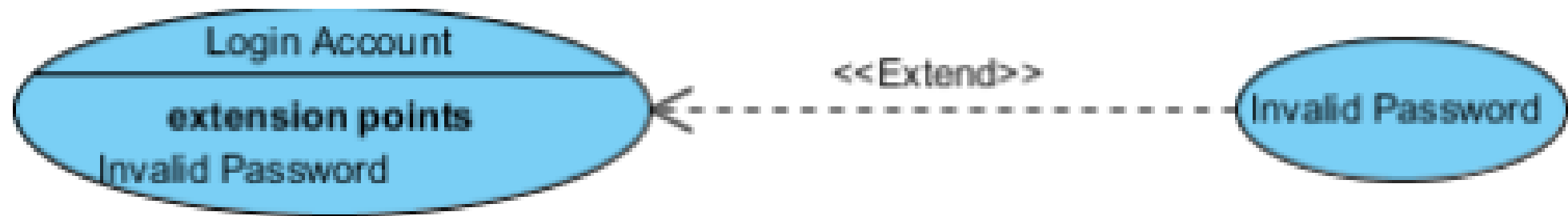
Include Relationship

- The Include Relationship indicates that a use case includes the functionality of another use case.
- It is denoted by a dashed arrow pointing **from the including use case to the included use case**. This relationship promotes modular and reusable design.

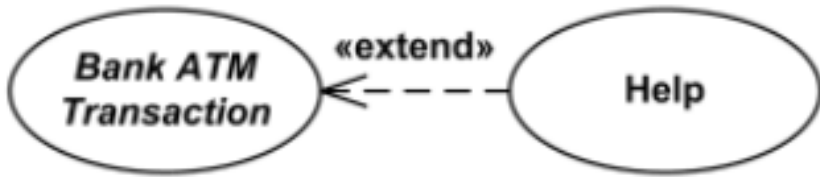
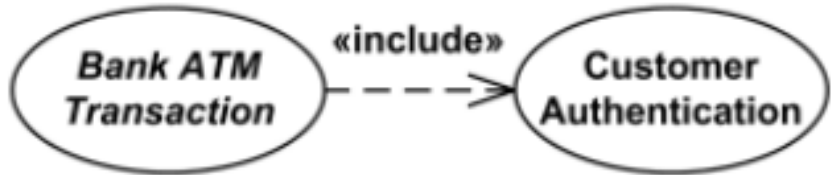


Extend Relationship

- The Extend Relationship illustrates that a use case can be extended by another use case under specific conditions. It is represented by a dashed arrow with the keyword “extend.” This relationship is useful for handling optional or exceptional behavior.

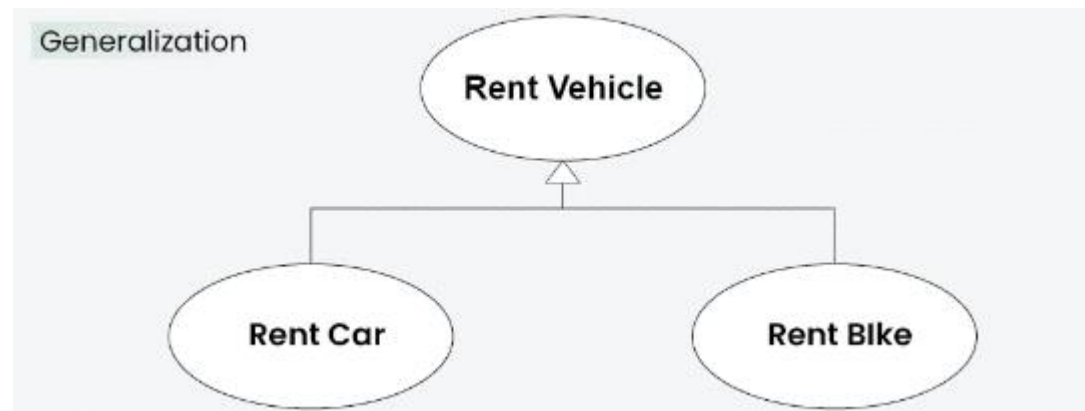


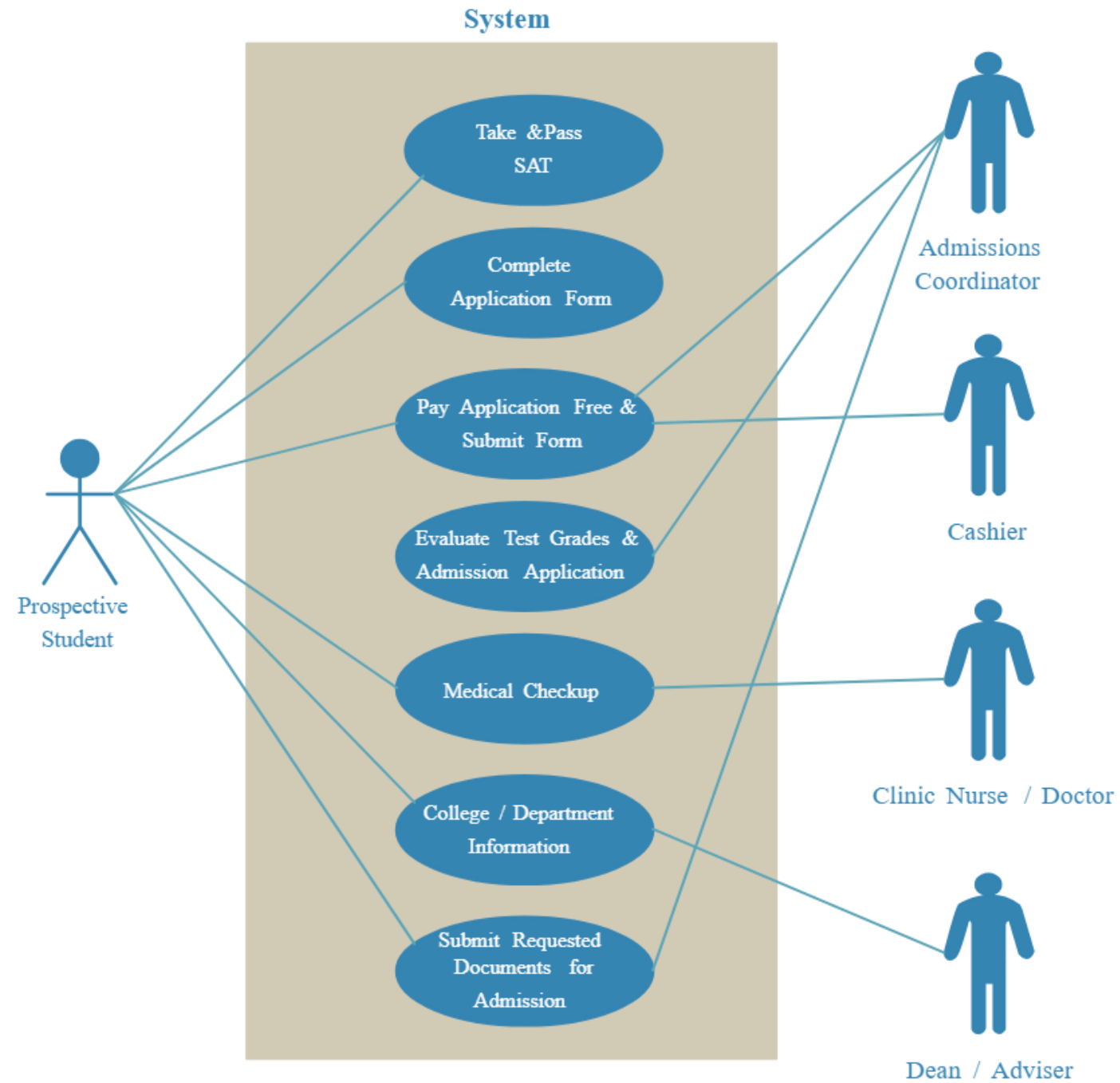
Include vs Extend

Extend	Include
	
Base use case is complete (concrete) by itself, defined independently.	Base use case is incomplete (abstract use case).
Extending use case is optional, supplementary.	Included use case required, not optional.
Has at least one explicit extension location.	No explicit inclusion location but is included at some location.
Could have optional extension condition.	No explicit inclusion condition.

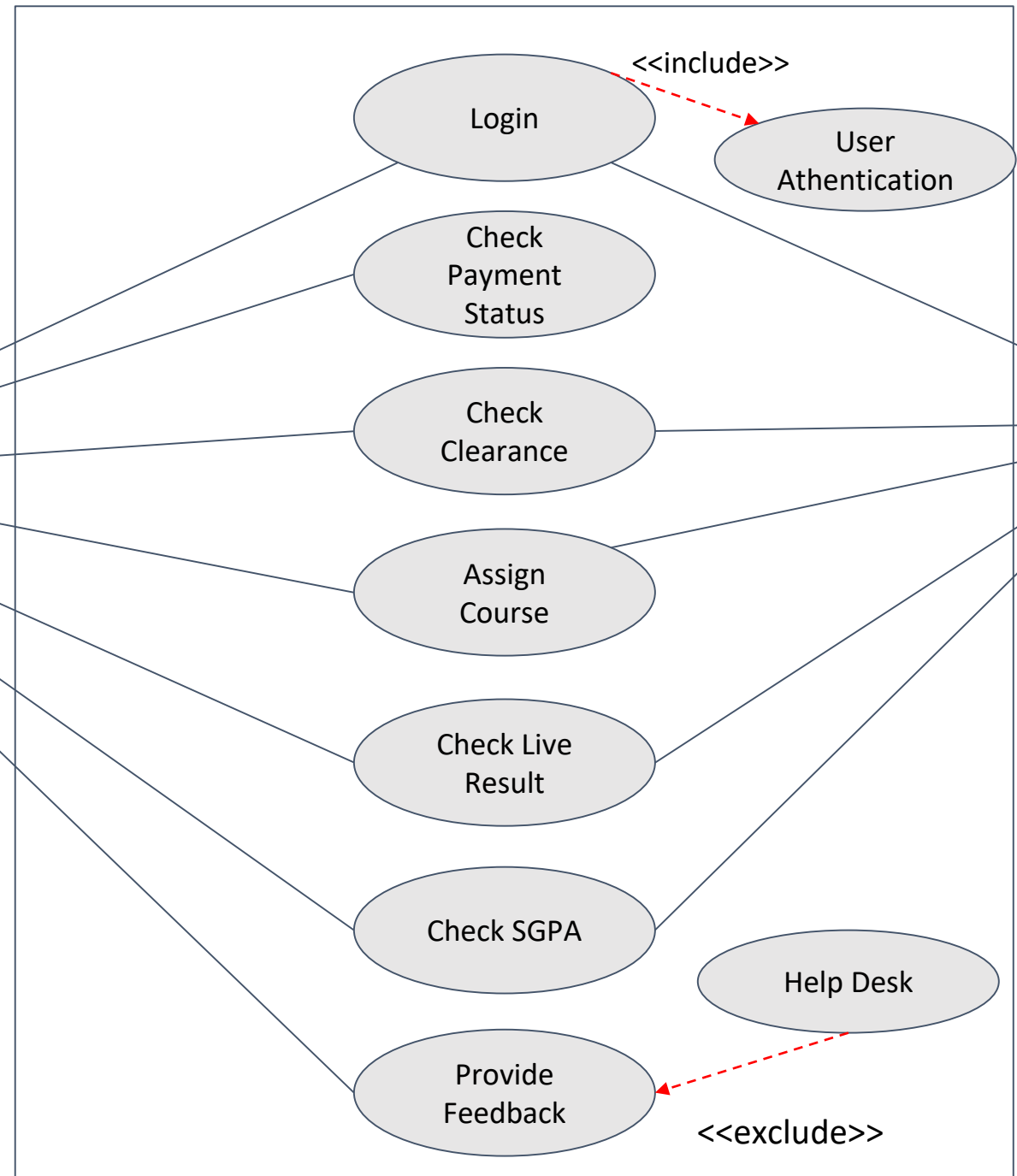
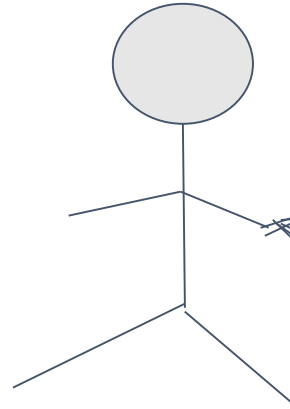
Generalization Relationship

- The Generalization Relationship establishes an “is-a” connection between two use cases, indicating that one use case is a specialized version of another.
- It is represented by an arrow pointing from the specialized use case to the general use case.





Student
Primary Actor



Teacher
Second Actor

